

Mechanical Engineer

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Department of Mechanical and Aerospace Engineering, Ohio State University, Columbus, OH, USA Fall 2006
Teaching Assistant, "Thermodynamics I"

- Contributed to designing experiments for a new thermodynamics laboratory.

ITCEN Co. (Industrial & Technical Consulting Engineers Company), Tehran, IRAN Mar. 2006–Sept. 2006
Senior Engineer

- Designed the layout of production lines for a pipe manufacturer.

Department of Mechanical Engineering, Sharif University of Technology, Tehran, IRAN Sept. 2003–Dec. 2005
Research Assistant

- Performed compression tests on bovine liver and characterized its material properties using the genetic algorithm and the finite element method. Developed an algorithm to obtain the optimal path initiation for the needle insertion into bovine liver for biopsy and brachytherapy purposes.

SAPCO Co. (Supplying Automotive Parts Company), Tehran, IRAN Summer 2001; Summer 2002
Intern

- Analyzed newly designed and produced automotive parts using mechanical tests such as Engine Test, Material Strength Test, etc.

HONORS AND AWARDS

MIT Outstanding Graduate Student Institute Award (2010). This award was given to the top two graduate students at the Department of Mechanical Engineering at MIT. The department has more than 500 graduate students; **NSF Fellowship for the GEM4-2010 program** (2010); **Highly Distinguished Student of University of Tehran** (1999–2003): A student who is in top 0.05% (out of ~500,000 applicants) in the nation-wide university entrance exam and his/her semester GPAs are above 17 out of 20.

JOURNAL PUBLICATIONS

Mech Eng et al., "Three-dimensional implicit hydrodynamic model of semi-flexible filaments", *in preparation*.

Mech Eng et al., "Diffusion coefficients of DNA origami structures", *in preparation*.

Mech Eng et al., "Brownian dynamics simulation of DNA origami structures", *in preparation*.

Mech Eng et al., "A finite element framework for Brownian dynamics simulation of proteins", *in preparation*.

Mech Eng, A. A. Fedorov, E. V. Fedorov, S. Ono, F. Matsumura, S. C. Almo, & M. Bathe, "Structure, evolutionary conservation, and conformational dynamics of Homo sapiens fascin-1, an F-actin crosslinking protein", *Journal of Molecular Biology*, 400 (2010), pp. 589-604.

Mech Eng, M. T. Ahmadian, & F. Janabi-Sharifi, "Modeling, simulation, and optimal initiation planning for needle insertion into the liver", *Journal of Biomechanical Engineering-Transactions of the ASME*, 132 (2010), p. 041001 (11 pages).

Mech Eng, M. Bathe, & K. J. Bathe, "The subspace iteration method in protein normal mode analysis", *Journal of Computational Chemistry*, 31 (2010), pp. 66-74.

M. T. Ahmadian, **Mech Eng**, & R. Abdollahpour, "A nonlinear viscoelastic modeling of brain and CSF deformation under tumor expansion", *International Journal of Scientific Research*, 16 (2006), pp. 425-428.

M. T. Ahmadian, **Mech Eng** R. Abdollahpour, S. Sharifi Sedeh, & K. Navi, "Application of car active suspension in vertical acceleration reduction of vehicle due to road excitation and its effect on human health", *International Journal of Scientific Research*, 16 (2006), pp. 429-434.

M. T. Ahmadian, R. Abdollahpour, & **Mech Eng**, "Effect of tumor location and its growth on stress distribution in the brain", *International Journal of Scientific Research*, 16 (2006), pp. 523-527.

OTHER PUBLICATIONS

3 first-author journal abstracts; 14 conference papers.

ACTIVITIES

- Sports: Soccer; table tennis; swimming; hiking; mountain climbing.
- Music: Singing.